Math 133: Quiz 6

Friday, October 23, 2026 10 points

Name: _____

Show your work and give exact answers. You may use differentiation rules. Work where each function is defined and differentiable. Answers may be left in factored form; you do not need to expand products or simplify quotients. A simple scientific calculator is allowed, but is not needed.

1. (7 points) Find each derivative.

   (a) $f(x) = 5x^3$.

   $f'(x) =$ _____

   (b) $f(x) = \dfrac{1}{x} - 6x$.

   $f'(x) =$ _____

   (c) $g(x) = (3x + 1)^4$.

   $g'(x) =$ _____

   (d) $h(x) = (x^2 + 1)e^x$.

   $h'(x) =$ _____

(e) $r(x) = \dfrac{x^2+3}{x-1}$.

$r'(x) =$ _____

(f) $s(x) = \ln(2x+1)$.

$s'(x) =$ _____

(g) $u(x) = \sqrt{3x-4}$.

$u'(x) =$ _____

2. (3 points) Find an equation of the tangent line to the graph of $y = \ln(x)$ at the point $(e, 1)$.

Tangent line: _____